

Ayush Kumar Singh

Data Engineer

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Professional Summary

Senior Data Engineer with **3.5+ years** building **real-time and batch** data infrastructure for high-volume **fintech platforms**. Expert in **CDC, Apache Kafka, AWS lake house (Hudi, Iceberg), PII/DPDP** compliance, and end-to-end **ETL pipelines**. Proven delivery across streaming systems, data governance, observability stacks, and **AI-driven pipeline automation**. Experienced in building **GenAI-powered data agents** (Microsoft **Fabric, Copilot**) with semantic modeling and row-level security Proficient in **Python, PySpark**, and a broad **AWS** services ecosystem.

Education

- Symbiosis Institute of Technology** **August 2019 - June 2023**
B. Tech in Computer Science and Engineering | **7.59 CGPA – Honours in Data Science**

Work Experience

- Bajaj Broking | Senior Data Engineer** **Aug 2025 - Present**

 - Built** and **published** an AI-grounded **Copilot agent** in Microsoft **Fabric** on a **Power BI semantic model** of client feature data, with **row-level security** restricting each **Relationship Manager** to their own clients, enabling self-serve **natural language Q&A** via Microsoft Teams.
 - Architected** and **maintained** scalable real-time data pipelines by implementing **Change Data Capture (CDC)** on **MS SQL Server** and streaming events through **Kafka**, reducing transactional database query load by **40–60%** and improving data availability for **downstream systems**.
 - Developed** and **deployed** an **AI-powered agent** on **AWS** to **monitor ETL pipelines**, automatically **detecting failures, data anomalies**, and performance **bottlenecks** to improve pipeline reliability.
- Bajaj Broking | Data Engineer** **Jul 2023 – Jul 2025**

 - Architected** scalable **AWS data lake** infrastructure using **Hudi, Iceberg**, and **Parquet** formats; applied encryption and governance controls to ensure **PII** compliance and enable secure, high-performance data access for analytics teams.
 - Engineered** a comprehensive monitoring solution for a trading application, incorporating **Grafana, Fluentd**, and **Elasticsearch**. Integrated **real-time logs and on-premises databases**, cutting report creation time by 80% and improving API performance by 40%.
 - Worked** with **cross-functional teams** to design reporting **data models, power real-time dashboards**, and support **AI-driven workflows** by transforming raw data into structured, governed datasets for analytics and decision-making.
 - Developed** a robust **ETL framework** for a **Customer Data Platform (CDP)** project using **AWS services (Glue, S3, Athena)**. Enhanced **data integration and processing workflows**, ensuring compliance with **data security standards** by **encrypting** and managing **Personally Identifiable Information (PII)**.
 - Led** the **integration** of vendor **data pipelines** into the **enterprise data lake house**, designing **scalable ingestion workflows** that transformed external datasets into **governed, analytics-ready tables** for **downstream reporting, analytics, and ML workloads**.
- Bajaj Broking | Data Engineer Intern** **Jan 2023 – Jun 2023**

 - Built a **Data Discovery** solution for identifying and classifying **PII** across multiple **databases**, including **names, phone numbers, emails, Aadhaar, and PAN**. Implemented encryption in compliance with Data Protection and Digital Privacy (DPDP) regulations.
 - Designed** and **implemented** real-time engagement systems for targeted campaigns, increasing campaign **CTR by 18–35%** and improving client **response rates by 33%**.
 - Engineered** an **Aadhaar masking** solution using **Easy OCR** and **Python**, improving data masking accuracy from **70%** to over **95%**.
 - Executed **RFM segmentation** models on trading data to power customer lifecycle analytics and targeted campaigns, driving a **12% increase** in **customer retention**.

Technical Skills

Languages: Python, PySpark, Go, Fluentd, R.
Database: MSSQL, MYSQL, PostgreSQL, MongoDB, Elasticsearch, Redis, Amazon Athena.
Tools: AWS, Git, NGINX, Grafana, AWS Services (S3, Glue, Lambda, API Gateway, SNS, Athena, KMS, IAM, Event Bridge).
Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, Keras, XGBoost, Deep Learning, AI Agents, Fabric Agent.
Data Engineering & Streaming: Apache Kafka, Change Data Capture (CDC), Real-time Streaming Pipelines, Batch Data Pipelines, Data Modeling.
Technologies/ Frameworks/ Libraries: Apache HUDI, Apache Iceberg, Apache Spark, Jupyter Notebooks, Docker, Parquet.
Soft Skills: Cross-functional Collaboration, Technical Problem Solving, Stakeholder Communication, Technical Leadership.

Projects

- AR/ Augmented Reality Quiz | Python, Flutter, Media Pipe Face mesh**
- Innovated** an **Augmented Reality Quiz** application using **Python, Flutter**, and **Media Pipe Face mesh**. The app leverages **face-tracking technology** to lock options based on the user's facial movements, enhancing **interactive** and **immersive** quiz experiences.

Achievements

- Super Heroes Award | Bajaj Broking**
- Recognised** for **consistently exceeding expectations and delivering exceptional value**, demonstrating outstanding performance and dedication.